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12 May 1970
SUPERSEDING
MIL-R-23489(SHIPS)
2 November 1962

MILITARY SPECIFICATION

RECEIVER, INFRARED (VIEWING SET), AN/SAR-7( )

1. SCOPE

1.1 This specification covers the design and construction of the Infrared Receiver AN/SAR-7( ), which is intended for use aboard ships at night, for observing near infrared illumination.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	-	Box, Fiberboard
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall

MILITARY

MIL-E-1/1049	-	Electron Tube, Image Converter Types 6914 and 6914A
MIL-B-18/20	-	Battery, Dry, BA-42
MIL-P-116	-	Preservation, Methods of
MIL-C-675	-	Coating of Glass Optical Elements (Antireflection)
MIL-S-901	-	Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment and Systems, Requirements for (Navy)
MIL-D-3464	-	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-Q-9858	-	Quality Program Requirements
MIL-F-14072	-	Finish for Ground Signal Equipment
MIL-E-16400	-	Electronic Equipment, Naval Ship and Shore, General Specification (Navy)
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	-	Definition of and Basic Requirements for Enclosure For Electric and Electronic Equipment
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-781	-	Reliability Tests Exponential Distribution
MIL-STD-785	-	Reliability Program for Systems and Equipment Development and Production

HANDBOOKS

MILITARY

MIL-HDBK-217	-	Reliability Stress and Failure Rate Data for Electronic Equipment

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois, 60606.)

NAVAIR 15-03-500
NAVSHIPS 0967-125-9010

(Application for copies should be addressed to the Naval Publications and Printing service, Building 157-3, Washington Navy Yard, Washington 20390 (code 074).)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3).

3.2 General requirements. The equipment shall be in accordance with the provisions of MIL-E-16400 except as hereinafter specified.

3.2.1 Temperature classification. The temperature classification of the equipment shall conform to class 2 of MIL-E-16400 except as hereinafter specified (see 4.5.4).

3.2.2 Power source. The equipment shall be designed to operate from not more than 3 volts direct current (dc) obtained from the enclosed battery.

3.2.3 Primary power line voltage and frequency. The primary power line voltage and frequency requirements are not applicable.

3.2.4 Service conditions. The equipment shall be constructed to give optimum performance and reliable service during continuous or intermittent periods of at least 6 months without the necessity for major adjustments or servicings. The equipment shall be capable of sustained operation in exposed positions under adverse weather conditions and shall withstand rough handling.

3.2.5 Warm-up. The equipment shall be fully operable within 15 seconds after the power is turned on.

3.2.6 Fungus nutrient or moisture sensitive materials shall not be used.

3.3 Reliability.

3.3.1 Mean-time-between-failure. The mean-time-between-failure (MTBF) for this equipment, excluding batteries shall be 4000 hours.

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3.3.2 Reliability program phase. The supplier shall establish and implement a reliability program in accordance with MIL-STD-785 (see 6.2.2).

3.3.3 Reliability predictions. The contractor shall perform a reliability prediction in accordance with the requirements of the paragraph "*How to make a Reliability Prediction Based on Stress Factors and Part Populations*" in MIL-HDBK-217 (see 6.2.2).

3.3.4 Failure rates. The primary source of failure rate data shall be MIL-HDBK-217. For those components not listed in MIL-HDBK-217, other sources of failure rate data may be used with the prior approval of the command or agency concerned.

3.4 Detail requirements.

3.4.1 Equipment composition. The equipment shall consist of a viewer, a carrying case, batteries, carrying straps and a nylon cover.

3.4.1.1 Viewer. The viewer shall consist of a main housing containing an infrared image converter tube, power supply and battery compartment, and one objective and two interchangeable ocular lens assemblies.

3.4.1.2 Shock. Parts shall be rigidly mounted to withstand the shock test specified in 4.5.8.

3.4.1.3 Dimensions and weight. The maximum dimensions and weight of the viewer shall be as follows:

(a) Dimension - 3-9/16 inches wide by 10-1/2 inches long by 4-5/8 inches high.
(b) Weight - 3.25 lbs. - All dimensions and weights shall be taken with the 8.6 power eyepiece set at plus 2 diopters and less batteries. Weights and dimensions are absolute maxima. Reduction of weight and size shall be a major design objective.

3.4.1.4 Time meters. Time meters are not required.

3.4.1.5 Synchros. Synchros are not required.

3.4.2 Optical.

3.4.2.1 Surface examination. The optical parts and the entire interior of the viewer shall be clean and free from oil, dust, dirt, or other foreign matter and free from any lubricant which might, in time, collect on any of the optical parts. The glass used for the optical elements shall be stable, nonhygroscopic, and free from strains, striae, bubbles, seeds or stones visible in the field of view or which would impair optical performance.

3.4.2.2 Optical surfaces. Optical surfaces, except reflecting and cemented surfaces, shall be coated with reflection reducing film in accordance with MIL-C-675.

3.4.2.3 The objective lens shall have a focal length of 3.5 inches providing a minimum field of view of 16 degrees. It shall be equivalent or superior to the f/2.2 Bausch and Lomb Cinephor and shall be corrected and coated for 0.90 micron radiation.

3.4.2.4 Eyepiece interchangeability. Two eyepieces or oculars, shall be provided with each equipment. They shall be mechanically interchangeable with respect to the central unit and shall be coated (see 3.4.2.2) for 0.550 micron radiation. One eyepiece shall have a free aperture of no less than 20 millimeters, and shall have a magnifying power of not less than 8.6 while the other shall be equivalent to that shown on figure 1. The 8.6 power ocular shall be variable in focus to accommodate the vision of individual operators. The larger ocular shall be fixed in focus.

3.4.2.5 Distortion. The overall distortion in the equipment shall be minimized. Field flattening lenses and distortion correcting eyepieces shall be provided for best performance. The image tube shall be enclosed in a mu-metal shield to minimize the electron-optical effects of magnetic fields.

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3.4.2.6 Ghost images. Ghost images or images of targets outside of the field of view shall not be permissible when the viewer is being held in the normal position with respect to the eye.

3.4.3 Mechanical. Both the objective and oculars shall be of the rotating focusing type.

3.4.3.1 The focus of the objective lens shall be adjustable from 50 feet to infinity while that of the oculars shall be adjustable over the range of +2 to -2 diopters. Thread tolerances between mated parts shall conform to Unified National Class 2 in accordance with Handbook H28 after any hardening, impregnation or preservative treatment with the further provision that there shall be sufficient lengths of thread in contact so that there shall be no side or end play between the optics assemblies and the main housing at any point within the focusing range. The unit shall not rattle when shaken.

3.4.3.1.1 The large ocular, equivalent to figure 1, shall be fixed at -1 diopter at the factory by means of a stop ring on the threaded portion of the viewer housing. With the stop ring in place and the large ocular replaced by the small 8.6 power ocular it shall be possible to focus the 8.6 power ocular over the range specified in 3.4.3.1. Optics shall possess sufficient focal length tolerance so that all units shall be individually interchangeable between viewers without the necessity for any adjustment other than to the ocular stop ring to accommodate different fixed focus oculars. The fixed and variable oculars shall not be furnished in matched sets.

3.4.3.1.2 Optical system interchangeability. The objective and ocular lens assemblies shall be interchangeable with corresponding parts of the Government furnished AN/SAR-7 or AN/SAR-7A Infrared Viewing Sets. Interchangeability applies to outside dimensions, mounting, focal length, field curvature, and functional performance.

3.4.3.1.2.1 Unless otherwise authorized by the command or agency concerned, lens assemblies shall conform to the dimensions specified in 3.4.3.1.2.1.1 through 3.4.3.1.2.1.3.

3.4.3.1.2.1.1 Objective lens assembly. The objective lens assembly shall be 3.5 inches focal length, 3-19/22 inches long, 2-9/32 inches major outside diameter on one end, and 2 inches minor outside diameter on the other end of the barrel. The external thread on the objective bezel and its mating thread which attaches the objective assembly to the main housing and serves as a focusing thread shall be 2,000-20 NS-3A and NS-2, respectively, in accordance with Handbook H28. Focal plane shall be 1.9 ±0.1 inches from the threaded end of the barrel.

3.4.3.1.2.1.2 Eyepiece assembly. Eyepiece assembly, long eye relief, 4-power, shall be 3-3/8 inches long, 4 inches major outside diameter on one end and 3-7/32 inches minor outside diameter on other end. Mountings shall be externally threaded on smaller end. Threads shall be 2.985 - 16 NS-2. Focal plane shall be 11.10 +0.8 -0.0) millimeters from threaded end of barrel.

3.4.3.1.2.1.3 Eyepiece assembly 8.6 power shall be 1-11/32 inches long, 1-13/16 inches in diameter. The 8.6 power eyepiece shall be assembled in the eyepiece mount which shall be interchangeable with the 4-power eyepiece. The eyepiece shall be attached by a 1.312-32 NS-2 external focusing thread in accordance with Handbook H28. The mating internal thread on the eyepiece mount shall be 1.312 NS-3B in accordance with Handbook H28. Focal plane shall be 18.42 ±.5 millimeters from end.

3.4.3.1.2.1.4 A heavy body silicone anti-seize compound shall be applied to screw threads used to assemble aluminum parts together including focusing threads.

3.4.3.2 Eyeguard cap. The small ocular shall be provided with a moulded black eyeguard cap of such shape that any cross-sections at right angles to the line of sight will be circular and concentric with the ocular axis. The size and shape of these caps shall be such as to allow the observer to see the entire field of view with the eye in a comfortable position, and exclude from the eye the major portion of light, except that transmitted through the ocular. The inner surfaces of the caps shall be ridged or corrugated to eliminate stray light. The caps shall be readily removable to facilitate cleaning the exit surface of the eye lens.

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3.4.3.3 The main bodies, or housings, shall be die castings, centrifugal castings, permanent mold castings, or extrusions of an aluminum base alloy, or shall be reinforced rigid structures of a corrosion resistant metal, such as Monel or titanium, or glass reinforced melamine or epoxy. Strength and rigidity with light weight shall be a major objective.

3.4.3.4 The insulator body, identified as part of assembly A-104 in NAVSHIPS 0967-125-9010 and associated manufacturing drawings shall be of General Electric Lexan polycarbonateresin or an equivalent material acceptable to the Command or Agency concerned in lieu of polystyrene.

3.4.4 Sensitivity. The viewer shall have a threshold sensitivity value of 0.10 nautical mile candles or better with the most favorable lens combination. The sensitivity shall be measured by a procedure equivalent to the following: A tungsten lamp, operated at a color temperature of 2850° Kelvin (K.) and filtered by a sheet of Polaroid XRX55 material is used as the source. A variable attenuator, composed of two convex first-surface mirrors, is used to reduce the intensity of the radiation incident on the viewer to the extent that the image of the source in the viewer cannot be seen by a fully dark adapted observer. The illumination which would then be incident on the viewer, were there no filter, multiplied by the filter factor (equivalent infrared transmission) expressed as nautical mile candles, and defines the sensitivity of the viewer. A nautical mile candle is by definition 2.7777×10^{-8} foot candles.

3.4.5 Resolving power. The minimum resolving power of the viewer shall be 2.5 minutes of arc in the center of the field and not less than 2.5 minutes of arc at any point in a 7.5 degree arc at the center of the field of view.

3.4.5.1 Corona and arcing. During the resolution test, the AN/SAR-7( ) unit shall be tested for freedom from arcing and corona (see 4.5.2.1).

3.4.6 Image tube. The infrared image converter tube shall conform to type 6914A of MIL-E-1/1049.

3.4.7 Power supply. The miniature power supply to convert low voltage dc required from batteries to the high voltage dc required for the operation of the type 6914A image converter tube shall be contained wholly within the housing of the viewer unit. Since the power supply is an integral part of the hand-held viewing unit the paramount design requirements are miniature size, minimum weight, low current drain, and long-term reliability.

3.4.7.1 The power supply shall be designed to operate from one type C-cell dry battery, MIL-Type BA-42 in accordance with MIL-B-18/20.

3.4.7.1.1 The power supply shall operate for 30 hours continuously on one BA-42 dry cell. Battery life shall be measured to that point where the power supply under a load of one microampere falls below 80 percent of the initial output voltage specified in 3.4.7.2.

3.4.7.2 The high voltage power supply shall be a single encapsulated unit with dimensions as follows:

Length overall - 2.718 ±0.02 inches
Diameter - 1.000 ±0.01 inches

It shall contain a transistor oscillator section, a transformer and a high voltage multiplier section containing silicon diodes and capacitors. The power supply shall have a maximum output voltage of 16,000 volts when fully charged batteries and a load equivalent to a dark type 6914A tube are used with the supply. The regulation shall be such that the output voltage is between 16,000 and 15,000 volts with any load between 1×10^{-9} amperes and 5×10^{-7} amperes, and shall be not less than 14,000 volts with a load of 1×10^{-6} amperes or less than 12,000 volts with a load of 2×10^{-6} amperes.

3.4.7.3 Battery compartment. The battery compartment shall be readily accessible for replacement of the battery. It shall be sealed from the interior of the main housing so that opening of the battery compartment does not impair the pressure tightness of the interior. The battery cap shall be fitted with an O-ring to make it watertight. The compartment shall be designed so that the battery can complete the circuit only when inserted with the correct polarity.

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3.4.7.4 Switch. A rotary type of ON-OFF waterproof switch shall be provided.

3.4.8 Mounting plate. The viewer shall be provided with a mounting plate or with grooves to fit the clamp for type C receivers as shown on figure 2. Instructions for making this clamp are included in the technical manual NAVSHIPS 0967-125-9010.

3.4.9 Carrying strap. The viewer shall be provided with a detachable binocular type strap for around-the-neck carry. The strap shall be of nylon or a similar moisture- and fungus-resistant material.

3.4.10 Protective cover. A protective cover shall be provided which may be fitted over the viewer when mounted on a searchlight. The cover shall be waterproofed dark gray or black nylon.

3.4.11 Construction details. Due to the high voltage present in this equipment, sharp corners, edges or burrs tending to cause corona shall be smoothed and rounded. Corona-inhibiting compound and potting techniques shall be used, as applicable, to reduce corona and to prevent leakage.

3.4.12 Safety features and circuit protection. This equipment shall be so designed and constructed as to eliminate the hazard of electrical shock to operating personnel. The circuit arrangement shall be designed so as to prevent damage to the equipment due to open or short circuit of the output terminals or failure of high voltage leads and connectors, and to protect transistors from transient high voltages. Safety interlocks are not required.

3.4.13 Blown fuse indicators. Blown fuse indicators are not required.

3.5 Enclosure. The main body of the viewer and the objective and large eyepieces shall be watertight to a minimum depth of 3 feet of water and airtight in accordance with MIL-STD-108 to a positive or negative pressure of 10 to 12 pounds per square inch (psi) referred to the interior. The housings of the main body and large eyepiece shall be fitted with valve arrangements for flushing and filling with dry nitrogen. Prior to delivery they shall be filled with dry nitrogen to a positive pressure of no less than 5 psi. The residual internal dew point shall be lower than -50 degrees centigrade (c) after filling.

3.6 Finish. The equipment and parts, shall be internally finished as follows: Aluminum shall be anodized and dyed black. Other metals shall be primed with a suitable metal primer and finished with a flat black enamel. The external priming coats shall be in accordance with MIL-E-16400. The finish coats shall be gray wrinkle enamel finished in accordance with type I of MIL-F-14072.

3.7 Carrying case. A carrying case of aluminum, fiberglass, or of moulded fungus resistant polyvinyl construction shall be provided. The case shall have a nondetachable hinged cover, a handle suitably placed for hand carry, and detachable strap for over-the-shoulder carry. The case shall accommodate and protect the viewing unit, the alternate optics, and one set of spare batteries. The units shall be securely fixed within the case when it is closed but easily removable when the case is opened. The case shall be watertight to 3 feet hydrostatic pressure and shall withstand the shock test of 4.5.8 without damage to the case or its contents.

3.8 Design approval. The design of the AN/SAR-7 equipment shall be submitted to the command or agency concerned for approval prior to delivery of a model for first article testing. Design data shall include all information necessary for a decision on the suitability of important features and shall include a complete set of shop drawings. Approval of the design will constitute acceptance of such features as size, form factor, arrangement, appearance, and general acceptability for service use. Design improvements are desirable and will be favorably considered. However, deviations from the detail requirements of this specification or of any referenced specification shall not be allowed unless duly and specifically approved in writing by the command or agency concerned. Approval of the design shall not relieve the contractor of the responsibility for meeting the performance requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as

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otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program requirement. The supplier shall provide and maintain a quality program acceptable to the Government for supplies and services covered by this specification. The quality program shall be in accordance with MIL-Q-9858.

4.1.1.1 Quality plan. A quality plan, applicable to the requirements of this specification shall be submitted to the Quality Assurance Representative for approval prior to first article inspection (see 6.2.2). If first article inspection is not required the quality plan shall be submitted for approval prior to quality conformance inspection. The quality plan shall reflect the following:

(a) Flow of material from inspection on receipt through all manufacturing processes to final shipment
(b) process and inspection points
(c) reference to process and inspection procedures
(d) differentiation between in plant inspections and subcontracted inspections for subassemblies and higher assemblies.

Any changes to the approved plan shall be submitted to the Quality Assurance Representative for approval prior use.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Test procedures. The supplier shall submit written test procedures including test equipment and tolerance limits to be used for first article inspection and quality conformance inspection. Each test shall be identified in accordance with 4.2 as to classification (see 6.2.2).

4.1.3.1 Procedures for inspections, tests, demonstrations, and so forth, shall contain step by step information which will enable the operator to properly perform the requirements of this specification. The method of examination shall be clearly defined.

4.1.3.2 The procedures shall contain a list of test equipment identified by name, model, range and accuracy. Special test equipment shall also be listed and identified.

4.1.3.3 Data sheets shall be included in the procedures. Conformance parameters shall be clearly indicated with tolerance limits. Whenever quantitative measurements are made, the result shall be shown by quantitative data.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)
    (1) Production inspection (see 4.4.1)
    (2) Production control inspection (see 4.4.2)
    (3) Environmental inspection (see 4.4.3)
    (4) Reliability testing (see 4.4.8)
(c) Inspection of preparation for delivery (see 4.6)

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4.3 First article inspection. Unless otherwise specified (see 6.2.1), two (2) units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table I.

TABLE I. First article inspection.

Examination or test	Requirement paragraph	Test method paragraph
Surface examination	3.4.2.1, 3.4.3.1, 3.4.3.1.2.1.4, 3.4.3.2, 3.4.6, 3.4.7.3, 3.4.8, 3.4.11, 3.6	4.5.1
Operating test	3.2.5, 3.4.2.4, 3.4.2.6, 3.4.4, 3.4.5, 3.4.5.1	4.5.2
Optical properties	3.4.2.2, 3.4.2.3, 3.4.2.4, 3.4.3.1, 3.4.3.1.1, 3.4.3.1.2.1.1, 3.4.3.1.2.1.2, 3.4.3.1.2.1.3	4.5.3
Dimensions and weights	3.4.1.3, 3.4.3.1, 3.4.3.1.2.1, 3.4.7.2	4.5.1
Heat test	3.2	4.5.4
Enclosure test	3.5, 3.7	4.5.5
Temperature test	3.2.1	4.5.4
Humidity test	3.2	4.5.4
Output voltage and current drain	3.4.7.2	4.5.6
Optics system interchangeability	3.4.3.1.1, 3.5.3.1.2	4.5.7
Electromagnetic interference emission and susceptibility and systems self-compatibility	3.2	4.5.1
Controls and control circuits	3.4.7.4	4.5.1
Shock test	3.4.11, 3.7	4.5.8
Vibration, inclination, and structural integrity	3.4.1.2, 3.4.11	4.5.1
Salt spray test	3.2	4.5.1
Magnetic test	3.2	4.5.1
Weld test	3.2	4.5.1
Battery life test	3.4.7.1	4.5.9

4.3.1 First article inspection report. A first article inspection report containing all the results of the first article inspection shall be prepared. The first article inspection report shall indicate specification compliance limits and contain data for all inspection and test results. The completed Form NAVEXOS 3373 as specified in MIL-S-901 for shock test and Resilient Mount Data Sheet as specified in MIL-E-16400 shall be included in the report along with evidence of the ability of the equipment to perform its intended function during and after shock and vibration tests (see 6.2.2). First article approval will not be granted unless the report is complete. Any production entered into prior to approval of the first article report will be at the suppliers risk.

4.4 Quality conformance inspection

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and

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reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with 4.1.3 and shall include the requirements of table II.

TABLE II. Production inspection

Examination or test	Requirement paragraph	Test method paragraph
Surface examination	3.4.2.1, 3.4.3.1, 3.4.3.1.2.1.4, 3.4.3.2, 3.4.6, 3.4.7.3, 3.4.8, 3.4.11, 3.6	4.5.1
Operating test	3.2.5, 3.4.2.4, 3.4.2.6, 3.4.4, 3.4.5, 3.4.5.1	4.5.2
Optical properties	3.4.2.2, 3.4.2.3, 3.4.2.4, 3.4.3.1, 3.4.3.1.1, 3.4.3.1.2.1.1, 3.4.3.1.2.1.2, 3.4.3.1.2.1.3	4.5.3

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to table III and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

TABLE III. Production control inspection

Examination or test	Requirement paragraph	Test method paragraph	AQL (Percent defective)
Dimensions and weight	3.4.1.3, 3.4.3.1, 3.4.3.1.2.1, 3.4.7.2	4.5.1	6.5
Heat test	3.2	4.5.4	10
Enclosure test	3.5, 3.7	4.5.5	6.5
Output voltage and current drain	3.4.7.2	4.5.6	6.5
Controls and control circuits	3.4.7.4	4.5.1	6.5
Optics system interchangeability	3.4.3.1.1, 3.4.3.1.2	4.5.7	10
Magnetic test	3.2	4.5.1	10
Weld test	3.2	4.5.1	10

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection. Environmental inspection shall be conducted to prove the durability of the equipment to withstand the effects of extremes of environment, service conditions, and hard usage. Environmental inspection shall be performed on equipment(s) that have passed production inspection as specified in 4.4.1 and shall include the tests listed in table IV.

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TABLE IV. Environmental inspection

Examination or test	Requirement paragraph	Test method paragraph
Temperature test	3.2.1	4.5.4
Humidity test	3.2	4.5.4
Shock test	3.4.1.2, 3.7	4.5.8
Battery life test	3.4.7.1	4.5.9
Electromagnetic interference emission and susceptibility and systems self-compatibility	3.2	4.5.1

4.4.3.1 Environmental inspection samples. Samples for the environmental inspection shall be selected without regard to their quality in accordance with the following:

(a) One sample from the first 25 production units fabricated.
(b) One sample from each successive 200 units produced, or fraction thereof.

4.4.4 Non-conformance and retest. If a sample unit fails a test specified in 4.4.2, 4.4.3 and 4.4.8 the supplier shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes, and so forth. The equipment shall then be retested.

4.4.5 Corrective action. If corrective action results in a change of part(s); electrical and mechanical layout or function; or if the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred immediately to the contracting officer. It shall then be determined by the command or agency concerned whether correction should be made to the equipments already delivered under the contract.

4.4.6 Reinspection of conforming production control and environmental test units. Unless otherwise specified, sample units which have been subjected to and passed production control and environmental inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

4.4.7 Quality conformance inspection report. The supplier shall submit to the Government inspecting activity prior to each shipment of the respective equipment a complete report detailing the results of all testing performed (see 6.2.2). The report shall list the serial numbers of equipments tested during the reporting period. During the conduct of tests, performance parameters shall be measured and verified in accordance with this specification, the report shall include completed Form NAVEXOS 3373 for the shock test and, when resilient mounts are used, compliance with figure 3 of MIL-E-16400. Equipment shall not be accepted unless the report is complete.

4.4.8 Reliability testing

4.4.8.1 Qualification testing. The first eight (8) production units shall be selected for testing in accordance with MIL-STD-781, test level D, test plan V1, except that voltage cycling is not required. Qualification testing is not required when there is no requirement for first article inspections.

4.4.8.2 Production testing. Every third month or every 300 equipments, which ever is more frequent, eight (8) equipments shall be randomly selected by the cognizant government inspector for testing in accordance with 4.4.8.1 herein.

4.4.8.3 Reliability test procedure. Reliability test procedure for qualification and production testing shall be in accordance with the "Detail Requirements" section of MIL-STD-781. Reports on reliability testing are required in accordance with the reliability test reports paragraph and subparagraphs of MIL-STD-781 (see 6.2.2).

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4.4.8.4 Disposition of tested equipment. Upon successful completion of reliability testing the equipment shall be subjected to the production inspection of 4.4.1 of this specification and may be accepted as part of the contract quantity provided it meets the requirements of this specification.

4.5 Inspection procedures.

4.5.1 Tests and examinations. Tests and examinations specified herein shall be conducted in accordance with the applicable inspection procedures of MIL-E-16400 except as follows:

4.5.2 Operating test. The equipment shall be energized and subjected to an operating test to insure the proper functioning and performance at an adequate number of points in the required range to demonstrate conformance to specification requirements, including safety. The operating test shall include, but not be limited to, the tests shown in table V.

TABLE V. Operating test

Examination or test	Requirement paragraph
Warm up	3.2.5
Ghost images	3.4.2.6
Sensitivity	3.4.4
Resolving power	3.4.5
Corona and arcing (see 4.5.2.1)	3.4.5.1
Eyepiece interchangeability	3.4.2.4

4.5.2.1 Corona and arcing test. Units shall be tested for freedom from arcing and corona. The arcing and corona test shall consist of observing the lapsed time between turning off of the unit and fading of the screen image. The unit shall be energized and focused on a resolution pattern for a period of not less than one minute. The unit shall then be switched off. The image of the resolution pattern shall remain on the screen of the image tube for at least 30 seconds. There shall be no visible or audible evidence of arcing, flashing, or flickering during the test.

4.5.3 Optical properties. The equipment shall be examined for optical surface coatings, focal length, field of view, free aperture, magnifying power, and length to focal plane for conformance to 3.4.2.2, 3.4.2.3, 3.4.2.4, 3.4.3.1, 3.4.3.1.1, 3.4.3.1.2.1.1, 3.4.3.1.2.1.2, 3.4.3.1.2.1.3.

4.5.4 Heat test, temperature test and humidity test. These tests shall be conducted in accordance with the applicable inspection procedures of MIL-E-16400. Battery failure during hot and cold operating tests shall not be cause for rejection of the equipment. Temperature limitations for tube operation shall be as specified in MIL-E-1/1049.

4.5.5 Enclosure test. In addition to the airtight and watertight tests of MIL-STD-108. The main body of the viewer and the objective and large (4 power) eyepiece assemblies shall be subjected to internal pressure and dew point tests for conformance to 3.5. The carrying case shall be subjected to the watertight test of MIL-STD-108 for conformance to 3.7.

4.5.6 Output voltage and current drain. The power supply shall be tested for output voltage and current drain for conformance with 3.4.7.2.

4.5.7 Optics system interchangeability. The objective and the 8.6 and 4 power eyepiece assemblies shall be individually interchanged with other viewer housings and with corresponding parts of AN/SAR-4, AN/SAR-7 and AN/SAR-7A equipment to demonstrate conformance with 3.4.3.1.1 and 3.4.3.1.2.

4.5.8 Shock test. Shock test shall be conducted on the viewer and the carrying case in accordance with the following:

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(a) Viewer. To demonstrate conformance to 3.4.11, the viewer shall be held with the eyepiece upward and dropped a distance of 6 feet into fine white sand 6 inches deep covered with a thin cloth. The viewer shall be undamaged and shall operate satisfactorily after the drop.

(b) Carrying case. To demonstrate conformance to 3.7, the carrying case with contents as specified herein shall be dropped a distance of three feet onto a hardwood floor. It shall be dropped once on each of four corners and once on the hinge side. The case and contents shall be undamaged after the five drops and the latches shall remain closed. The viewer shall operate satisfactorily after the test.

4.5.9 Battery life test. One BA-42 dry cell shall be installed into the power supply and operated continuously for 30 hours to obtain a measurement of the battery life in accordance with 3.4.7.1.

4.6 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2.1).

5.1.1 Level A. Each Infrared Viewing Set shall be packaged in accordance with method IIb of MIL-P-116. Dessiccant shall conform to MIL-D-3464, type II or III. Technical literature shall be packaged in a waterproof, heat sealed, transparent bag, minimum 4 mil, and shall not be placed within the barrier used to package any other item. Loose fill polystyrene shall not be used for packaging. Adhesives shall not be used in the closure of wraps, bags or boxes. Items comprising a complete set shall be placed in a consolidated package conforming to PPP-B-636, class weather resistant.

5.1.1.1 Other items. Other items, such as equipment repair parts, shall be individually packaged in transparent packages in accordance with MIL-E-17555.

5.1.2 Level C. Each complete set shall be packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2.1).

5.2.1 Level A. Sets packaged as specified in 5.1 shall be packed in containers conforming to PPP-B-636, V grade, special requirements for type 2 load, or PPP-B-640, class 2. Closure and reinforcement shall be in accordance with the applicable box specification except that boxes shall be reinforced with nonmetalic tape banding. Use of loose fill polystyrene and adhesives shall be as required in 5.1.1.

5.2.2 Level B. Sets packaged as specified in 5.1 shall be packed in containers conforming to PPP-B-636, special requirements or PPP-B-640. Closure, reinforcement and use of loose fill polystyrene and adhesives shall be as required in 5.2.1.

5.2.3 Level C. Sets packaged as specified in 5.1 shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers packing or method of shipment shall comply with Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by NAVAIR 15-03-500 or the contract or order, unit packages, and shipping containers shall be marked in accordance with MIL-STD-129, except that all exterior level A and B pack markings shall be waterproof.

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6. NOTES

6.1 Intended use. The equipment covered in this specification is intended for use in observing near infrared illumination aboard ships at night.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).
(c) Number of first article samples to be submitted if other than specified in 4.3.

6.2.2 Contract data requirements. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data required by this specification includes, but is not restricted to the following:

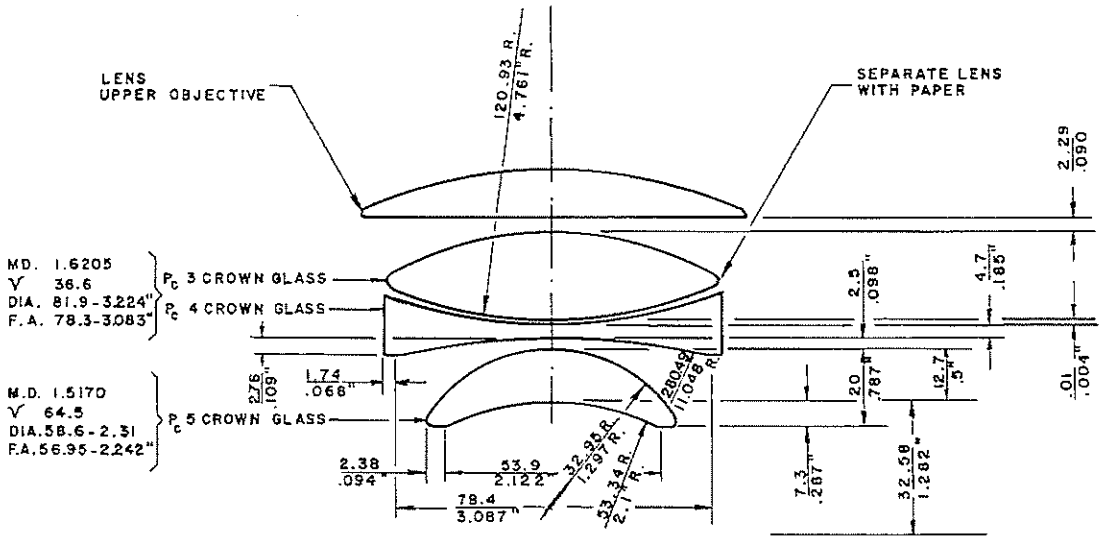
(a) Reliability program phase (see 3.3.2)
(b) Reliability predictions (see 3.3.3)
(c) Quality plan (see 4.1.1.1)
(d) Test procedures (see 4.1.3)
(e) First article inspection report (see 4.3.1)
(f) Quality conformance inspection report (see 4.4.7)
(g) Reliability test procedure reports (see 4.4.8.3)

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the aid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - EC

(Project 5850-N025)

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COMPLETE OBJ.

B.F.L. = 2.000
E.F.L. = 68.881
F.F.L. = 110.318

OBJ. WITHOUT P_c 6

B.F.L. = 38.776
E.F.L. = 66.450
F.F.L. = 59.853

EC 435

Figure 1. Large aperture eyepiece AN/SAR-7

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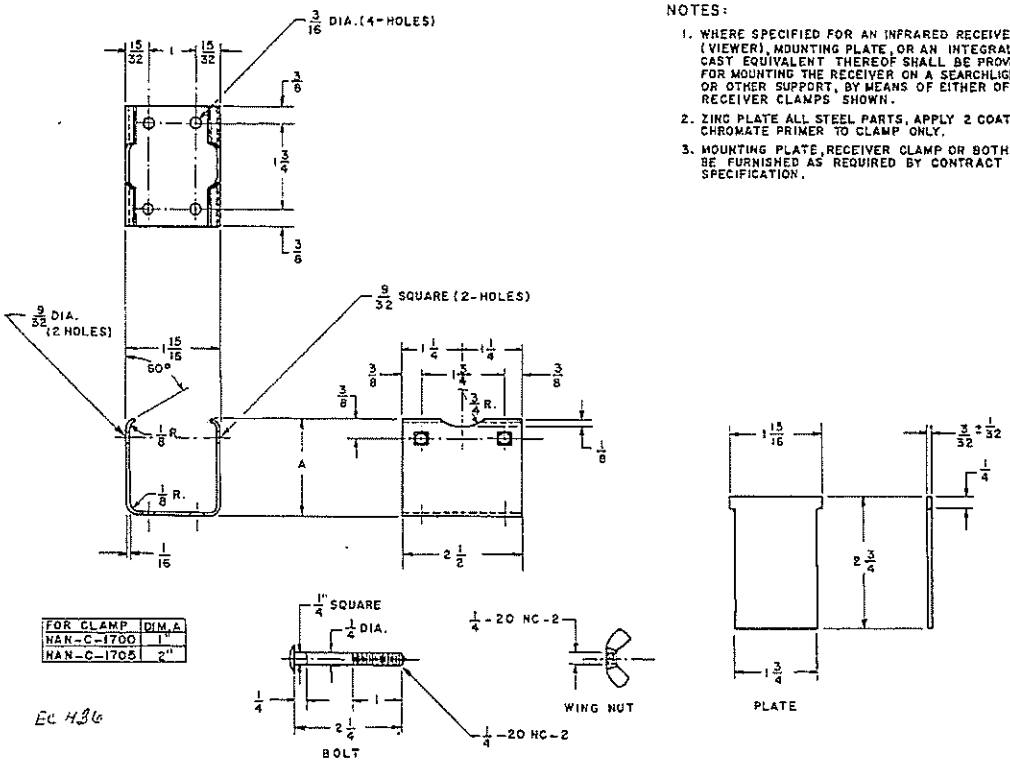


Figure 2. Mounting plate for infrared receivers.

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